# Supplementary materials of "D3I-COMCF: dissimilarity-driven dual-interaction via co-occurrence motifs and complementary fragments for drug-drug interaction prediction"

## I. Evaluation metrics

ACC, AUC, F1-score and AP are defined as follows:

$$\begin{aligned}
&ACC = \frac{TP+TN}{TP+FP+FN+TN},\\
&AUC = \frac{1}{|\mathcal{P}|\cdot|\mathcal{N}|}\sum_{p\in\mathcal{P}}\sum_{n\in\mathcal{N}}\mathbb{I}\left(\text{score}(p)>\text{score}(n)\right)+\frac{1}{2|\mathcal{P}|\cdot|\mathcal{N}|}\sum_{p\in\mathcal{P}}\sum_{n\in\mathcal{N}}\mathbb{I}\left(\text{score}(p)=\text{score}(n)\right),\\
&F1-score = \frac{2TP}{FN+FP+2TP},\\
&AP = \sum_{k=1}^{z}(R_k-R_{k-1})\cdot P_k,
\end{aligned} \tag{1}$$

where *TP* indicates the number of positive samples predicted as positive; *FP* is the number of negative samples predicted as positive; *FN* is the number of positive samples predicted as negative; *TN* represents the number of negative samples predicted as negative; $p$ indicates an arbitrary positive sample in the positive sample set $\mathcal{P}$; $n$ represents an arbitrary negative sample in the negative sample set $\mathcal{N}$; $\mathbb{I}(\cdot)$ is the indicator function, takes value 1 if the condition inside holds, 0 otherwise; score$(p)$ and score$(n)$ represent model prediction score of positive sample $p$ and negative sample $n$, respectively; $z$ is the total number of samples; $k$ is the sorting threshold index, corresponding to the top $k$ samples sorted by prediction score (descending); $R_k$ indicates the Recall at the $k$-th sorting threshold (ratio of true positives in top $k$ samples to total true positives); $P_k$ indicates the Precision at the $k$-th sorting threshold (ratio of true positives in top $k$ samples to the number of top $k$ samples)

## II. Supplementary description

### A. Motif identifier and vocabulary construction

**Motif identifier (canonical key).** For each BRICS-derived and merged clique $C$ (motif candidate) with $k$ atoms, we define a permutation-invariant motif key by combining (1) the multiset of atom types and (2) the multiset of intra-clique bond types. Atom types are represented by atomic numbers $\{Z_i\}_{i\in C}$, and we define:

$$A(C) = \mathrm{sort}\big(\{Z_i\}_{i\in C}\big), \tag{2}$$

where $\mathrm{sort}(\cdot)$ removes dependence on the atom enumeration order within $C$. For bond types, we enumerate all atom pairs $(i,j)$ with $i<j$ and assign $b_{ij}=1,2,3,4$ for single, double, triple, and aromatic bonds, respectively. If no direct covalent bond exists between $i$ and $j$, we set $b_{ij}=-1$. We then define:

$$B(C) = \mathrm{sort}\big(\{b_{ij}\}_{i<j,\ i,j\in C}\big). \tag{3}$$

where $B(C)$ is the sorted multiset of bond-type codes over all atom pairs in $C$, making it invariant to atom ordering. Finally, the canonical motif key is $K(C)=(A(C),B(C))$, which is mapped to a unique integer motif ID via a global vocabulary: an existing key reuses its ID; otherwise, a new ID is assigned sequentially.

The three examples in Table S1 illustrate how $K(C)$ distinguishes motifs by jointly considering atom types and intra-clique bond types. The first two cliques share the same atom-type multiset and thus the same $A(C)=(6,6,8)$. However, changing the C–C bond from single to double changes the bond-code multiset from $B(C)=(-1,1,1)$ to $B(C)=(-1,1,2)$, resulting in different keys and motif IDs. In contrast, the first and third cliques share the same bond-code multiset $B(C)=(-1,1,1)$ but have different atom-type multisets ($A(C)=(6,6,8)$ and $(6,7,8)$), and are therefore treated as distinct motifs. Notably, each example contains a non-bonded atom pair encoded as $-1$ (consistent with the definition of $b_{ij}$). Moreover, since both $A(C)$ and $B(C)$ are sorted multisets, $K(C)$ is invariant to atom enumeration order, ensuring reproducible motif identification.

TABLE S1: Three illustrative examples of canonical motif identifiers.

| Clique atoms | Atomic numbers $A(C)$ | Pairwise bonds | Bond codes $B(C)$ | Motif key $K(C)$ |
|---|---|---|---|---|
| $\{C, C, O\}$ | $(6, 6, 8)$ | $\{C{-}C,\ C{-}O,\ C\cdots O\}$ | $(-1, 1, 1)$ | $((6, 6, 8), (-1, 1, 1))$ |
| $\{C, C, O\}$ | $(6, 6, 8)$ | $\{C{=}C,\ C{-}O,\ C\cdots O\}$ | $(-1, 1, 2)$ | $((6, 6, 8), (-1, 1, 2))$ |
| $\{C, N, O\}$ | $(6, 7, 8)$ | $\{C{-}N,\ N{-}O,\ C\cdots O\}$ | $(-1, 1, 1)$ | $((6, 7, 8), (-1, 1, 1))$ |

**Note.** In the 'Pairwise bonds' column, '$X{-}Y$' denotes a single bond, '$X{=}Y$' denotes a double bond, and '$X\cdots Y$' indicates that no direct covalent bond exists between the atom pair within the clique (thus $b_{ij} = -1$). The bond codes $B(C)$ are reported as the sorted multiset of $\{b_{ij}\}$, consistent with $B(C) = \mathrm{sort}(\{b_{ij}\})$.

### *B. Negative sampling protocols*

Let $\mathcal{D}$ denote the drug set and $\mathcal{P}$ denote the set of known positive DDI triplets in the corresponding benchmark split. A positive triplet is denoted as $(d_h, d_t, r)$, where $d_h$ and $d_t$ are the head and tail drugs and $r$ is the DDI type.

**Random-Neg protocol.** Following the standard corruption-based benchmark protocol, for each positive triplet $(d_h, d_t, r)$, we generate one negative triplet by randomly replacing either the head drug or the tail drug. If the head drug is selected for corruption, a replacement drug $d'_h$ is uniformly sampled from the candidate drug set, and the corrupted triplet becomes $(d'_h, d_t, r)$. If the tail drug is selected for corruption, the corrupted triplet becomes $(d_h, d'_t, r)$. **The generated triplet is accepted only if it is not included in the known positive set $\mathcal{P}$ under the benchmark annotation.** Otherwise, the replacement drug is re-sampled. This procedure keeps the positive-to-negative ratio at 1:1. We emphasize that such generated negatives are benchmark-defined negatives, i.e., drug pairs or triplets not annotated as known positives in the benchmark, rather than experimentally verified non-interacting drug pairs.

For example, given a known positive triplet $(d_1, d_2, r)$, Random-Neg may randomly choose to corrupt the head drug and sample $d_5$ from $\mathcal{D}$. If $(d_5, d_2, r) \notin \mathcal{P}$, then $(d_5, d_2, r)$ is used as a benchmark-defined negative triplet. Alternatively, if the tail drug is selected for corruption, a sampled drug $d_8$ can yield $(d_1, d_8, r)$, provided that this triplet is not annotated as a known positive.

**Hard-Neg@Top10 protocol.** To construct more challenging structure-aware negative samples, we additionally introduce Hard-Neg@Top10. Before evaluation, we compute Morgan fingerprint Tanimoto similarities between drugs and build a top-10 structurally similar candidate list $\mathcal{N}_{10}(d)$ for each drug $d$. For each positive triplet $(d_h, d_t, r)$, the corruption side is selected in the same way as in the Random-Neg protocol. **However, instead of uniformly sampling the replacement drug from the full candidate drug set, the replacement drug is preferentially selected from the top-10 structurally similar candidates of the corrupted drug.** Specifically, if the head drug is corrupted, the replacement drug $d'_h$ is selected from $\mathcal{N}_{10}(d_h)$, and the generated triplet $(d'_h, d_t, r)$ is accepted only if it is not annotated as a known positive in $\mathcal{P}$. If the tail drug is corrupted, $d'_t$ is selected from $\mathcal{N}_{10}(d_t)$ and $(d_h, d'_t, r)$ is accepted under the same filtering rule. If no valid replacement can be found from the top-10 candidate list after filtering known positives, the procedure falls back to the standard Random-Neg sampling.

For example, given a positive triplet $(d_1, d_2, r)$, suppose the head drug $d_1$ is selected for corruption and its top-10 structurally similar candidates are $\mathcal{N}_{10}(d_1) = \{d_3, d_4, \ldots, d_{12}\}$. Hard-Neg@Top10 preferentially selects a replacement drug from this list. If $(d_3, d_2, r) \notin \mathcal{P}$, then $(d_3, d_2, r)$ is used as a hard-negative triplet. If all top-10 candidates lead to triplets already annotated as known positives, the method falls back to random corruption. Therefore, Hard-Neg@Top10 increases the structural similarity between the original drug selected for corruption and its replacement drug, while still following the benchmark requirement of avoiding known positive triplets.

It should be noted that Hard-Neg@Top10 is used as a complementary structure-aware stress test rather than as a replacement for the standard benchmark protocol. **Since the generated hard negatives are candidate triplets not annotated as known positives, they should not be interpreted as experimentally confirmed non-DDI cases.** Instead, this protocol evaluates whether a model can better control false positives when the corrupted drug is structurally similar to the original drug.

### *C. Motif construction variants and additional splitting rules*

To clarify the motif granularity controlled by BRICS decomposition, additional splitting rules, and intra-drug clique merging, we define three motif construction variants: BRICS-only, BRICS+splitting, and BRICS+splitting+merging. The last one is used as the default setting of D3I-COMCF.

Given a molecule graph $G = (V, E)$, the BRICS-only variant first identifies BRICS bonds using RDKit and removes these bonds from $G$. The connected components of the remaining atom graph are then used as motif candidates. This setting preserves relatively large BRICS fragments and does not apply any additional splitting or overlap-based merging.

In the BRICS+splitting variant, we start from local covalent-bond cliques and then apply BRICS bond breaking. For each BRICS bond $(u, v)$, the corresponding two-atom clique is removed and the two endpoint atoms are inserted as singleton cliques. We further apply two auxiliary splitting rules. First, for a two-atom clique crossing a ring/non-ring boundary, the clique is split so that the non-ring-side atom is represented as a singleton clique. Second, for a high-degree non-ring atom, defined as a non-ring atom with more than two neighbors, its incident two-atom cliques are removed and the central atom and its neighboring atoms are inserted as singleton cliques. This setting keeps the additional splitting rules but does not perform the final intra-drug overlap-based clique merging. Exact duplicate cliques are removed, while overlapping but non-identical cliques are retained. Therefore, BRICS+splitting is used to examine the effect of over-fragmented motif granularity.

In the BRICS+splitting+merging variant, we apply the same BRICS decomposition and additional splitting rules as above, followed by intra-drug clique merging. Specifically, if two cliques from the same molecule share at least one atom, they are iteratively merged into their union:

$$C_i \leftarrow C_i \cup C_j, \quad \text{if } C_i \cap C_j \neq \varnothing. \tag{4}$$

This merging process continues until no overlapping clique pair remains within the molecule. This default setting is designed to reduce redundant over-fragmentation caused by auxiliary splitting and to recover more chemically coherent motif units.

TABLE S2: Atom (node) feature used in the SMG (55D)

| Feature name | Possible values / definition | Dimension |
|---|---|---|
| Atom symbol (one-hot) | {C, N, O, S, F, Si, P, Cl, Br, Mg, Na, Ca, Fe, As, Al, I, B, V, K, Tl, Yb, Sb, Sn, Ag, Pd, Co, Se, Ti, Zn, H, Li, Ge, Cu, Au, Ni, Cd, In, Mn, Zr, Cr, Pt, Hg, Pb, Unknown} | 44 |
| Atom degree | $\mathrm{GetDegree}/10$ | 1 |
| Implicit valence | $\mathrm{get\_atom\_valence(atom)}$ | 1 |
| Formal charge | $\mathrm{GetFormalCharge}$ | 1 |
| Radical electrons | $\mathrm{GetNumRadicalElectrons}$ | 1 |
| Hybridization (one-hot) | {SP, SP2, SP3, SP3D, SP3D2} | 5 |
| Aromatic flag | $\mathrm{GetIsAromatic} \in \{0, 1\}$ | 1 |
| Total number of Hs | $\mathrm{GetTotalNumHs}$ | 1 |

TABLE S3: Bond (edge) feature schema used in the SMG (16D)

| Feature name | Possible values | Dimension |
|---|---|---|
| Bond type (multi-hot) | {SINGLE, DOUBLE, TRIPLE, AROMATIC, IONIC} | 5 |
| Conjugation (one-hot) | $\mathrm{GetIsConjugated} \in \{0, 1\}$ or nonbond as an extra category | 3 |
| Ring membership (one-hot) | $\mathrm{IsInRing} \in \{0, 1\}$ or nonbond as an extra category | 3 |
| Stereo (one-hot) | {STEREONONE, STEREOANY, STEREOZ, STEREOE, nonbond} | 5 |

## III. Supplementary data

### A. *Supplementary table*

TABLE S4: Details of four datasets

| Dataset | Number of DDI types | Number of drugs | Number of training samples | Construction method | Mechanism of action | Belonging paradigm |
|---|---|---|---|---|---|---|
| DB | 86 | 1706 | 122,756 | Based on DrugBank 5.0 database, all drug pairs with interactions and their corresponding unique interaction types are reserved. | Pharmacokinetics | Multi-class |
| Twosides | 963 | 645 | 2,928,823 | Based on Twosides database (from clinical side effect reports (FAERS)), and following the data processing standards of Zitnik et al., the types of low-frequency DDI (the specified frequency of occurrence is less than 500 times) were excluded. | Pharmacodynamics | Multi-label |
| DeepDDI | 2 | 1704 | 215,037 | - | - | Binary |
| ZhangDDI | 2 | 544 | 76,518 | DDI records were integrated from several databases such as ChEMBL, KEGG, and DrugBank. | - | Binary |

**Note.** Among the three folds used in the experiment, the number of test samples in S1 ranges from 6,719 to 6,935, and the number of test samples in S2 ranges from 58,739 to 59,562, and their training set sizes are close to those in DB.

TABLE S5: Summary of no-MCS cases in four datasets

| Dataset | Number of drug pairs in the training and validation sets (deduplicated by drug pair) | Number of drug pairs with no MCS found (or MCS timeout) | Proportion |
|---|---|---|---|
| DB | 153,187 | 1,332 | 0.0087 |
| Twosides | 63,443 | 1,308 | 0.0206 |
| DeepDDI | 115,044 | 1,028 | 0.0089 |
| ZhangDDI | 36,006 | 51 | 0.0014 |

TABLE S6: Hyperparameters and training configuration

| Item | Setting |
|---|---|
| Hidden dimension | 128 |
| RBF encoding dimension | 32 |
| Attention heads (MM-LDR) | 2 |
| Attention heads (AF-LDR) | 2 |
| Optimizer | Adam |
| Base learning rate | 0.001 |
| MBGAT learning rate | 0.0005 |
| Max epochs | 200 |
| Early stopping patience | 30 epochs (no validation improvement) |
| Evaluation protocol | 3-fold cross-validation (mean $\pm$ std) |

TABLE S7: Details of CMMIG for four datasets

| Information \ Dataset | DB | Twosides | DeepDDI | ZhangDDI |
|---|---|---|---|---|
| Molecule node counts | 1,706 | 645 | 1,704 | 544 |
| Motif node counts | 542 | 305 | 543 | 276 |
| Edge counts | 8,524 | 3,220 | 8,534 | 2,691 |
| Drug pairs sharing $\geq$ 1 motif | 1,336,782 | 189,327 | 1,333,331 | 136,604 |
| Motifs shared by $\geq$ 2 drugs | 365 | 149 | 264 | 130 |
| Avg shared motifs per connected pair | 2.2649 | 2.2356 | 2.2683 | 2.1958 |
| Max shared motifs between one drug pair | 12 | 9 | 12 | 7 |
| Self-loop counts | 10 | 5 | 16 | 0 |

**Note.** 'Drug pairs sharing $\geq$1 motif' refers to the number of unique drug pairs that share at least one motif. For each motif, all drugs containing that motif are collected and all pairwise drug combinations are enumerated; each drug pair is counted only once, regardless of how many motifs it shares. 'Motifs shared by $\geq$2 drugs' refers to the number of motifs appearing in at least two distinct drugs. 'Avg shared motifs per connected pair' refers to the average number of shared motifs over all drug pairs sharing at least one motif. 'Self-loops' refers to the number of drugs from which no motif is ultimately extracted.

TABLE S8: Hardware configuration and training cost

| Dataset | GPU | VRAM | RAM | #Params | Notes |
|---|---|---|---|---|---|
| Twosides | NVIDIA GeForce RTX 3090 | 24 GB | 32 GB | $\approx 6.47 \times 10^7$ | peak ∼22 GB/epoch; ∼24,000 s/epoch |
| DB | NVIDIA GeForce RTX 3060 | 12 GB | 16 GB | $\approx 7.24 \times 10^6$ | –; ∼ 1,431 s/epoch |
| DeepDDI | NVIDIA GeForce RTX 3060 | 12 GB | 16 GB | $\approx 1.74 \times 10^6$ | –; ∼ 1,379 s/epoch |
| ZhangDDI | NVIDIA GeForce RTX 3060 | 12 GB | 16 GB | $\approx 1.74 \times 10^6$ | –; ∼ 330 s/epoch |

TABLE S9: Efficiency and computational cost comparison between the original Train-only protocol and its Wo_MCS ablation during the training (RTX 3060 12 GB).

| Metric | Train-only(MCS+OHNE) | Wo_MCS (Fully Connected) |
|---|---|---|
| Epoch time (s) | 1217.63 | 173.96 |
| Train avg cross-edge build time per pair (ms) | 53.4783 | 0.0685 |
| Val avg cross-edge build time per pair (ms) | 47.2182 | 0.0675 |
| Train loader wait total (s) | 891.81 | 1.42 |
| Val loader wait total (s) | 181.01 | 3.51 |
| Train avg cross edges per pair | 58.07 | 684.41 |
| Val avg cross edges per pair | 65.70 | 733.97 |
| Train total pair-graph edges | 48,675,462 | 344,227,384 |
| Train avg edges per instance | 218.46 | 1513.55 |
| Train peak GPU allocated (MB) | 3576.18 | 6752.45 |
| Train peak GPU reserved (MB) | 6834.00 | 11718.00 |
| Val peak GPU allocated (MB) | 954.94 | 3024.94 |
| Val peak GPU reserved (MB) | 8526.00 | 11014.00 |

TABLE S10: Efficiency and computational cost comparison between the original Train-only protocol and its Wo_MCS ablation under the S1 setting (new drug, new drug) (RTX 3060 12 GB).

| **Metric** | **Train-only (MCS+OHNE)** | **Wo_MCS (Fully Connected)** |
|---|---|---|
| Whole eval loop time (s) | 178.0523 | 9.2680 |
| DataLoader total wait time (s) | 175.2326 | 5.2153 |
| Model forward total time (s) | 2.3088 | 3.6272 |
| Avg DataLoader per batch (s) | 35.0465 | 1.0431 |
| Avg forward per batch (s) | 0.4618 | 0.7254 |
| Avg batch total time (s) | 35.5083 | 1.7685 |
| Peak GPU allocated (MB) | 1634.26 | 5861.42 |
| Peak GPU reserved (MB) | 2726.00 | 11190.00 |

TABLE S11: Efficiency and computational cost comparison between the original Train-only protocol and its Wo_MCS ablation under the S2 setting (new drug, existing drug) (RTX 3060 12 GB).

| **Metric** | **Train-only(MCS+OHNE)** | **Wo_MCS (Fully Connected)** |
|---|---|---|
| Whole eval loop time (s) | 637.1608 | 38.2432 |
| DataLoader total wait time (s) | 618.8863 | 10.6340 |
| Model forward total time (s) | 17.8213 | 27.2395 |
| Avg DataLoader per batch (s) | 15.4722 | 0.2659 |
| Avg forward per batch (s) | 0.4455 | 0.6810 |
| Avg batch total time (s) | 15.9177 | 0.9468 |
| Peak GPU allocated (MB) | 1695.03 | 6043.06 |
| Peak GPU reserved (MB) | 6098.00 | 11150.00 |

TABLE S12: Efficiency and computational cost comparison between the original Transductive scenario (RTX 3060 12 GB) and its Wo_MCS ablation (RTX 3060 12 GB) during training.

| **Metric** | **Transductive (MCS+OHNE)** | **Wo_MCS (Fully connected)** |
|---|---|---|
| Epoch time (s) | 1468.96 | 114.02 |
| Train loader wait total (s) | 1032.52 | 6.14 |
| Train forward+backward total (s) | 117.93 | 88.15 |
| Val loader wait total (s) | 309.75 | 10.63 |
| Val forward total (s) | 8.30 | 8.60 |
| Peak GPU allocated (MB) | 7574.81 | 13290.60 |
| Peak GPU reserved (MB) | 11822.00 | 19796.00 |

TABLE S13: When randomly shuffling test labels while keeping predicted scores unchanged, the AUC values (%) obtained by S1 and S2 under the Train-only protocol

| Fold \ Setting | S1 | S2 |
|---|---|---|
| fold0 | 49.66 | 50.17 |
| fold1 | 49.95 | 50.02 |
| fold2 | 48.96 | 49.89 |

TABLE S14: When randomly shuffling test labels while keeping predicted scores unchanged, the AP values (%) obtained by S1 and S2 under the Train-only protocol

| Fold \ Setting | S1 | S2 |
|---|---|---|
| fold0 | 50.06 | 50.09 |
| fold1 | 50.15 | 50.05 |
| fold2 | 49.15 | 49.94 |

TABLE S15: Number of positive test samples for the representative DDI types shown in the S1 fold-wise comparison under the Train-only protocol.

| DDI type (rel_id) | Fold 0 | Fold 1 | Fold 2 |
|---|---|---|---|
| 3 | 99 | 254 | 158 |
| 31 | 54 | 39 | 52 |
| 5 | 153 | 75 | 189 |
| 29 | 41 | 43 | 13 |
| 46 | 1553 | 1428 | 1503 |
| 48 | 2349 | 2140 | 2333 |
| 72 | 1025 | 774 | 743 |
| 59 | 288 | 328 | 300 |

TABLE S16: Number of positive test samples for the representative DDI types shown in the S2 fold-wise comparison under the Train-only protocol.

| DDI type (rel_id) | Fold 0 | Fold 1 | Fold 2 |
|---|---|---|---|
| 54 | 22 | 39 | 34 |
| 14 | 42 | 46 | 42 |
| 60 | 145 | 156 | 96 |
| 68 | 76 | 81 | 69 |
| 3 | 1352 | 1763 | 1475 |
| 15 | 1713 | 1063 | 1538 |
| 46 | 11616 | 11219 | 11330 |
| 48 | 19228 | 18516 | 19218 |

TABLE S17: D3I-COMCF's performance results of all folds in Train-only protocol on DB(%).

| | S1 (new drug, new drug) | | | | | | S2(new drug, existing drug) | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Method / Metric | ACC | AUC | F1-score | AP | Precision | Recall | ACC | AUC | F1-score | AP | Precision | Recall |
| Fold 0 | 98.93 | 100.00 | 98.92 | 100.00 | 100.00 | 97.86 | 99.41 | 100.00 | 99.41 | 100.00 | 100.00 | 98.82 |
| Fold 1 | 93.36 | 100.00 | 92.89 | 100.00 | 100.00 | 86.72 | 95.56 | 100.00 | 95.36 | 100.00 | 100.00 | 91.13 |
| Fold 2 | 99.23 | 100.00 | 99.23 | 100.00 | 100.00 | 98.47 | 99.90 | 100.00 | 99.90 | 100.00 | 100.00 | 99.81 |
| Fold 3 | 99.29 | 100.00 | 99.29 | 100.00 | 100.00 | 98.59 | 99.78 | 100.00 | 99.78 | 100.00 | 100.00 | 99.56 |

**Note.** For the DSN-DDI baseline (3-fold cross-validation was employed), results under the cold-start scenario were available for four folds. However, in the main text, the mean and standard deviation were computed using only the first three folds, while the Supplementary Material presents the values of all six evaluation metrics for all four folds.

TABLE S18: Sensitivity analysis of the RBF encoding dimension $K$ (and the corresponding edge feature dimension) on representative settings (%).

| Setting | $K$ | ACC | AUC | F1-score | AP | Precision | Recall |
|---|---|---|---|---|---|---|---|
| Train-only S1 | 32 | 97.17 ± 3.31 | 100.00 ± 0.00 | 97.01 ± 3.57 | 100.00 ± 0.00 | 100.00 ± 0.00 | 94.35 ± 6.61 |
| Train-only S1 | 64 | 97.77 ± 3.07 | 100.00 ± 0.01 | 97.65 ± 3.27 | 100.00 ± 0.01 | 99.98 ± 0.04 | 95.56 ± 6.15 |
| Train-only S2 | 32 | 98.29 ± 2.38 | 100.00 ± 0.00 | 98.22 ± 2.49 | 100.00 ± 0.00 | 100.00 ± 0.00 | 96.59 ± 4.75 |
| Train-only S2 | 64 | 98.88 ± 1.03 | 100.00 ± 0.00 | 98.86 ± 1.05 | 100.00 ± 0.00 | 100.00 ± 0.01 | 97.76 ± 2.07 |
| DeepDDI | 32 | 95.63 ± 0.16 | 99.10 ± 0.07 | 97.06 ± 0.11 | 99.67 ± 0.02 | 98.68 ± 0.08 | 95.49 ± 0.20 |
| DeepDDI | 64 | 94.90 ± 0.45 | 98.96 ± 0.11 | 96.56 ± 0.30 | 99.61 ± 0.04 | 98.66 ± 0.17 | 94.54 ± 0.43 |

**Note.** $K = 32$ was adopted in the main experiments because it provided the most consistent performance across datasets and protocols. Although $K = 64$ improved the Train-only protocol on DB, it did not yield consistent gains on the binary benchmarks.

TABLE S19: Ablation study on whether to use RBF-encoded edge weights in the MBGAT layer on DeepDDI (%).

| Model | ACC | AUC | F1-score | AP | Precision | Recall |
|---|---|---|---|---|---|---|
| D3I-COMCF | 95.63 ± 0.16 | 99.10 ± 0.07 | 97.06 ± 0.11 | 99.67 ± 0.02 | 98.68 ± 0.08 | 95.49 ± 0.20 |
| Wo_RBF_edge_encoding | 95.06 ± 0.14 | 98.95 ± 0.09 | 96.67 ± 0.09 | 99.61 ± 0.03 | 98.56 ± 0.13 | 94.84 ± 0.07 |

**Note.** In Wo_RBF_edge_encoding, the CMMIG construction and scalar edge weights are kept unchanged, while the scalar-to-RBF edge encoding in the MBGAT layer is removed. The original scalar edge weight is directly used as a one-dimensional edge feature in GATConv.

TABLE S20: Ablation study on whether to use rare-motif-aware weighting in CMMIG on DeepDDI (%).

| Model | ACC | AUC | F1-score | AP | Precision | Recall |
|---|---|---|---|---|---|---|
| D3I-COMCF | 95.63 ± 0.16 | 99.10 ± 0.07 | 97.06 ± 0.11 | 99.67 ± 0.02 | 98.68 ± 0.08 | 95.49 ± 0.20 |
| Uniform molecule-motif weights | 95.09 ± 0.27 | 98.87 ± 0.15 | 96.69 ± 0.18 | 99.57 ± 0.06 | 98.46 ± 0.20 | 94.99 ± 0.31 |

**Note.** In uniform molecule-motif weights, the merged CMMIG structure, motif definition, and motif features are kept unchanged, while the original TF-IDF-style molecule-motif edge weights are replaced with uniform weights (all set to 1.0). This ablation is designed to isolate the contribution of the rare-motif-aware weighting mechanism itself.

TABLE S21: Controlled comparison of cross-molecular connection strategies on DeepDDI (%).

| Model | ACC | AUC | F1-score | AP | Precision | Recall |
|---|---|---|---|---|---|---|
| D3I-COMCF | 95.63 ± 0.16 | 99.10 ± 0.07 | 97.06 ± 0.11 | 99.67 ± 0.02 | 98.68 ± 0.08 | 95.49 ± 0.20 |
| A3 (MCS only, Wo_OHNE) | 94.78 ± 0.30 | 98.93 ± 0.09 | 96.47 ± 0.21 | 99.60 ± 0.03 | 98.57 ± 0.05 | 94.47 ± 0.35 |
| A4 (Random sparse cross edges, edge-budget matched) | 95.22 ± 0.58 | 99.06 ± 0.10 | 96.77 ± 0.40 | 99.65 ± 0.04 | 98.73 ± 0.07 | 94.89 ± 0.74 |

**Note.** A3 removes OHNE and retains only the direct MCS mapping edges, in order to isolate the contribution of complementary-fragment expansion beyond shared-backbone alignment. A4 replaces the original MCS+OHNE cross-molecular edges with random sparse cross edges while strictly matching the edge budget of the original construction for each drug pair. The larger variance of A4 is expected because the sampled cross-molecular edges are random, even under the same edge-budget constraint.

TABLE S22: Additional stricter cold-start stress tests on the **S1** setting (new drug, new drug) on DB (%).

| Setting | ACC | AUC | F1-score | AP | Prec | Rec |
|---|---|---|---|---|---|---|
| Train-only | 97.17 ± 3.31 | 100.00 ± 0.00 | 97.01 ± 3.57 | 100.00 ± 0.00 | 100.00 ± 0.00 | 94.35 ± 6.61 |
| A5 (hop = 1) | 98.05 ± 3.11 | 100.00 ± 0.00 | 97.94 ± 3.30 | 100.00 ± 0.00 | 100.00 ± 0.00 | 96.10 ± 6.23 |
| A5 (hop = 2) | 97.17 ± 3.31 | 100.00 ± 0.00 | 97.01 ± 3.57 | 100.00 ± 0.00 | 100.00 ± 0.00 | 94.35 ± 6.61 |
| A6 (stats-from-subset, 50%) | 88.87 ± 11.44 | 99.98 ± 0.03 | 86.20 ± 14.91 | 99.98 ± 0.03 | 100.00 ± 0.00 | 77.73 ± 22.88 |

**Note.** A5 reuses the trained Train-only model weights and only restricts the visible CMMIG context at test time to a local $h$-hop subgraph around the current batch anchors. A6 is retrained and reevaluated with motif-frequency/statistical quantities estimated from only 50% of the training drugs.

TABLE S23: Additional stricter cold-start stress tests on the **S2** setting (new drug, existing drug) on DB (%).

| Setting | ACC | AUC | F1-score | AP | Prec | Rec |
|---|---|---|---|---|---|---|
| Train-only | 98.29 ± 2.38 | 100.00 ± 0.00 | 98.22 ± 2.49 | 100.00 ± 0.00 | 100.00 ± 0.00 | 96.59 ± 4.75 |
| A5 (hop = 1) | 98.52 ± 2.41 | 100.00 ± 0.00 | 98.46 ± 2.52 | 100.00 ± 0.00 | 100.00 ± 0.00 | 97.04 ± 4.82 |
| A5 (hop = 2) | 98.29 ± 2.38 | 100.00 ± 0.00 | 98.22 ± 2.49 | 100.00 ± 0.00 | 100.00 ± 0.00 | 96.59 ± 4.75 |
| A6 (stats-from-subset, 50%) | 97.20 ± 2.42 | 100.00 ± 0.00 | 97.08 ± 2.53 | 100.00 ± 0.00 | 99.99 ± 0.01 | 94.41 ± 4.85 |

**Note.** A5 reuses the trained Train-only model weights and only restricts the visible CMMIG context at test time to a local $h$-hop subgraph around the current batch anchors. A6 is retrained and reevaluated with motif-frequency/statistical quantities estimated from only 50% of the training drugs.

TABLE S24: Increased-ratio Random-Neg evaluation under the Trans setting (%).

| Setting | ACC | AUC | F1-score | AP | Precision | Recall |
|---|---|---|---|---|---|---|
| D3I-COMCF | 99.99 ± 0.01 | 100.00 ± 0.00 | 99.99 ± 0.01 | 100.00 ± 0.00 | 99.99 ± 0.01 | 100.00 ± 0.00 |
| Test5 | 99.92 ± 0.12 | 100.00 ± 0.00 | 99.75 ± 0.37 | 100.00 ± 0.00 | 99.93 ± 0.07 | 99.58 ± 0.73 |
| Test10 | 99.96 ± 0.07 | 100.00 ± 0.00 | 99.76 ± 0.39 | 100.00 ± 0.00 | 99.94 ± 0.05 | 99.57 ± 0.73 |

**Note.** Test5 and Test10 denote relation-constrained corruption-based Random-Neg evaluation at test time, where each positive instance is paired with 5 or 10 randomly generated benchmark-defined negatives, respectively, while the original training procedure remains unchanged. These settings increase the random-negative ratio but do not use structure-aware hard negatives.

TABLE S25: Increased-ratio Random-Neg evaluation under the Train-only protocol on **S1** (%).

| Setting | ACC | AUC | F1-score | AP | Precision | Recall |
|---|---|---|---|---|---|---|
| Train-only | 97.17 ± 3.31 | 100.00 ± 0.00 | 97.01 ± 3.57 | 100.00 ± 0.00 | 100.00 ± 0.00 | 94.35 ± 6.61 |
| Test5 | 99.06 ± 1.06 | 100.00 ± 0.00 | 97.01 ± 3.42 | 100.00 ± 0.00 | 100.00 ± 0.00 | 94.35 ± 6.30 |
| Test10 | 99.25 ± 0.53 | 100.00 ± 0.00 | 95.60 ± 3.22 | 99.98 ± 0.02 | 99.99 ± 0.01 | 91.70 ± 5.83 |

TABLE S26: Increased-ratio Random-Neg evaluation under the Train-only protocol on **S2** (%).

| Setting | ACC | AUC | F1-score | AP | Precision | Recall |
|---|---|---|---|---|---|---|
| Train-only | 98.29 ± 2.38 | 100.00 ± 0.00 | 98.22 ± 2.49 | 100.00 ± 0.00 | 100.00 ± 0.00 | 96.59 ± 4.75 |
| Test5 | 99.43 ± 0.79 | 100.00 ± 0.00 | 98.22 ± 2.50 | 100.00 ± 0.00 | 100.00 ± 0.00 | 96.58 ± 4.75 |
| Test10 | 99.50 ± 0.60 | 99.99 ± 0.01 | 97.08 ± 3.57 | 99.98 ± 0.04 | 100.00 ± 0.00 | 94.47 ± 6.62 |

TABLE S27: Label-shuffling sanity check under increased-ratio Random-Neg evaluation in the Train-only protocol.

| Setting | S1 | | S2 | |
|---|---|---|---|---|
| | AUC | AP | AUC | AP |
| Test5 shuffled | 0.4999 ± 0.03 | 0.1672 ± 0.11 | 0.5003 ± 0.09 | 0.1666 ± 0.05 |
| Test10 shuffled | 0.4974 ± 0.12 | 0.0905 ± 0.06 | 0.4991 ± 0.05 | 0.0906 ± 0.02 |

**Note.** Under label shuffling, AUC approaches 0.5 and AP approaches class prevalence, confirming that the near-saturated AP/AUC under the true labels is not caused by leakage or evaluation artifacts.

TABLE S28: Train-only retrained with a 1:5 Random-Neg training ratio: evaluation on **S1** (%).

| Setting | ACC | AUC | F1-score | AP | Precision | Recall |
|---|---|---|---|---|---|---|
| Original Train-only | 97.17 ± 3.31 | 100.00 ± 0.00 | 97.01 ± 3.57 | 100.00 ± 0.00 | 100.00 ± 0.00 | 94.35 ± 6.61 |
| Retrained (train neg = 5), test neg = 5 | 97.63 ± 1.85 | 99.79 ± 0.30 | 92.14 ± 6.41 | 99.35 ± 0.92 | 99.90 ± 0.15 | 85.82 ± 10.95 |
| Retrained (train neg = 5), test neg = 1 | 91.84 ± 6.97 | 96.58 ± 4.76 | 91.15 ± 7.79 | 97.18 ± 3.92 | 97.41 ± 3.64 | 85.82 ± 10.95 |

TABLE S29: Train-only retrained with a 1:5 Random-Neg training ratio: evaluation on **S2** (%).

| Setting | ACC | AUC | F1-score | AP | Precision | Recall |
|---|---|---|---|---|---|---|
| Original Train-only | 98.29 ± 2.38 | 100.00 ± 0.00 | 98.22 ± 2.49 | 100.00 ± 0.00 | 100.00 ± 0.00 | 96.59 ± 4.75 |
| Retrained (train neg = 5), test neg = 5 | 98.54 ± 0.89 | 99.94 ± 0.09 | 95.38 ± 2.90 | 99.77 ± 0.31 | 99.93 ± 0.11 | 91.30 ± 5.23 |
| Retrained (train neg = 5), test neg = 1 | 94.75 ± 3.85 | 98.32 ± 2.36 | 94.52 ± 4.08 | 98.70 ± 1.83 | 98.19 ± 2.54 | 91.14 ± 5.40 |

TABLE S30: Sensitivity analysis of AF-LDR depth on the S1 setting under the Train-only protocol (%).

| AF-LDR blocks | ACC | AUC | F1-score | AP | Precision | Recall |
|---|---|---|---|---|---|---|
| 3 (default) | 97.17 ± 3.31 | 100.00 ± 0.00 | 97.01 ± 3.57 | 100.00 ± 0.00 | 100.00 ± 0.00 | 94.35 ± 6.61 |
| 2 | 94.86 ± 4.62 | 99.98 ± 0.03 | 94.42 ± 5.04 | 99.98 ± 0.03 | 100.00 ± 0.00 | 89.73 ± 9.23 |
| 4 | 96.87 ± 3.15 | 100.00 ± 0.00 | 97.06 ± 2.23 | 100.00 ± 0.00 | 100.00 ± 0.00 | 94.34 ± 4.21 |

TABLE S31: Sensitivity analysis of AF-LDR depth on the S2 setting under the Train-only protocol (%).

| AF-LDR blocks | ACC | AUC | F1-score | AP | Precision | Recall |
|---|---|---|---|---|---|---|
| 3 (default) | 98.29 ± 2.38 | 100.00 ± 0.00 | 98.22 ± 2.49 | 100.00 ± 0.00 | 100.00 ± 0.00 | 96.59 ± 4.75 |
| 2 | 98.30 ± 1.89 | 100.00 ± 0.00 | 98.24 ± 1.96 | 100.00 ± 0.00 | 100.00 ± 0.00 | 96.60 ± 3.77 |
| 4 | 98.31 ± 1.55 | 100.00 ± 0.00 | 98.04 ± 1.52 | 100.00 ± 0.00 | 100.00 ± 0.00 | 96.18 ± 2.92 |

TABLE S32: Computational cost comparison between 3-block and 4-block AF-LDR settings under the Train-only protocol. Results were recorded on the same fold, the same epoch, and the same RTX 3060 12 GB GPU.

| Metric | 3 blocks (default) | 4 blocks |
|---|---|---|
| Train loader wait total (s) | 902.3559 | 812.9346 |
| Train step total (s) | 131.0019 | 200.2396 |
| Avg train loader wait / batch (s) | 2.0696 | 1.8645 |
| Avg train step / batch (s) | 0.3005 | 0.4593 |
| Val loader wait total (s) | 150.2138 | 155.2645 |
| Val forward total (s) | 3.9538 | 4.7394 |
| Avg val loader wait / batch (s) | 8.8361 | 9.1332 |
| Avg val forward / batch (s) | 0.2326 | 0.2788 |
| Train pair-graph total edges (pos+neg) | 48,349,436 | 48,853,574 |
| Train avg edges per instance | 216.9985 | 219.2611 |
| Val pair-graph total edges (pos+neg) | 5,423,398 | 5,355,002 |
| Val avg edges per instance | 219.0564 | 216.2938 |
| Train peak GPU allocated (MB) | 3535.42 | 4562.48 |
| Train peak GPU reserved (MB) | 7746.00 | 7168.00 |
| Val peak GPU allocated (MB) | 962.29 | 985.16 |
| Val peak GPU reserved (MB) | 9034.00 | 8456.00 |

**Note.** The 2-block variant is omitted from the cost table because its predictive performance is already clearly inferior to the default 3-block setting. DataLoader waiting time is more affected by the host-side data pipeline and runtime fluctuation, whereas train/validation step time and GPU memory usage more directly reflect model-side computational cost.

TABLE S33: Complexity and computational cost of D3I-COMCF and representative variants under the warm-start scenario on DB.

| Model | Params (M) | Epoch time (s) | Train loader/MCS (s) | Train fwd+bwd (s) | Val loader/MCS (s) | Val fwd (s) | Peak alloc. (GB) | Peak reserved (GB) |
|---|---|---|---|---|---|---|---|---|
| Wo_MCS | 7.24 | 112.72 | 5.05 | 89.23 | 9.22 | 8.74 | 13.14 | 19.32 |
| Wo_CO | 6.35 | 1412.37 | 1038.21 | 44.63 | 325.50 | 3.50 | 3.08 | 9.85 |
| Wo_MM | 2.66 | 1363.74 | 958.27 | 95.08 | 302.67 | 7.19 | 6.15 | 11.48 |
| Wo_AF | 1.82 | 28.15 | 2.08 | 20.48 | 4.31 | 0.83 | 0.56 | 0.78 |
| D3I-COMCF | 7.24 | 1494.80 | 1044.25 | 115.49 | 326.62 | 7.90 | 6.58 | 11.27 |

**Note.** Params denotes the number of trainable parameters. 'fwd' and 'bwd' denote forward and backward propagation, respectively. 'alloc.' denotes peak allocated GPU memory, and 'reserved' denotes peak CUDA reserved memory reported by PyTorch. Train loader/MCS and Val loader/MCS include data loading and MCS+OHNE-based pair-graph construction time. Val fwd denotes the forward time on the validation set, which is used as the held-out inference cost. All measurements were obtained on fold 0 at the epoch achieving the best validation performance under the same hardware and batch-size settings.

TABLE S34: Complexity and computational cost of D3I-COMCF and representative variants under the cold-start scenario on DB.

| Model | Params (M) | Epoch time (s) | Train loader/MCS (s) | Train fwd+bwd (s) | Val loader/MCS (s) | Val fwd (s) | Peak alloc. (GB) | Peak reserved (GB) |
|---|---|---|---|---|---|---|---|---|
| Wo_MCS | 7.24 | 168.61 | 2.18 | 155.30 | 2.94 | 5.72 | 6.80 | 11.34 |
| Wo_CO | 6.35 | 1157.30 | 929.32 | 69.60 | 154.17 | 1.63 | 1.71 | 4.36 |
| Wo_MM | 2.66 | 1163.00 | 863.57 | 137.05 | 156.39 | 3.45 | 3.24 | 8.07 |
| Wo_AF | 1.82 | 44.60 | 1.23 | 38.91 | 1.79 | 0.71 | 0.38 | 0.54 |
| Train-only | 7.24 | 1217.63 | 891.81 | 138.47 | 181.01 | 3.71 | 3.49 | 8.33 |

**Note.** All measurements were obtained on fold 0 at the epoch achieving the best validation performance under the same hardware and batch-size settings. Peak GPU memory is affected by the batch size, the number of drug pairs processed per batch, and the size of the constructed pair graphs. Therefore, the memory values are intended to compare variants under the same protocol rather than to directly compare warm-start and cold-start protocols.

TABLE S35: Performance comparison between the standard random-negative protocol and the structure-based Hard-Neg@Top10 protocol on DB under the Train-only cold-start setting (%). Results are reported as mean $\pm$ standard deviation over three folds.

| Setting | Negative protocol | ACC | AUC | F1-score | AP | Precision | Recall |
|---|---|---|---|---|---|---|---|
| S1 | Random-Neg | 97.15 $\pm$ 3.34 | 100.00 $\pm$ 0.00 | 96.99 $\pm$ 3.61 | 100.00 $\pm$ 0.00 | 100.00 $\pm$ 0.00 | 94.31 $\pm$ 6.67 |
| S1 | Hard-Neg@Top10 | 51.14 $\pm$ 0.96 | 58.77 $\pm$ 8.97 | 65.82 $\pm$ 2.06 | 57.93 $\pm$ 7.57 | 50.59 $\pm$ 0.50 | 94.31 $\pm$ 6.67 |
| S2 | Random-Neg | 98.31 $\pm$ 2.34 | 100.00 $\pm$ 0.00 | 98.24 $\pm$ 2.46 | 100.00 $\pm$ 0.00 | 100.00 $\pm$ 0.00 | 96.62 $\pm$ 4.69 |
| S2 | Hard-Neg@Top10 | 52.80 $\pm$ 1.26 | 72.44 $\pm$ 11.90 | 67.16 $\pm$ 1.23 | 71.49 $\pm$ 13.13 | 51.50 $\pm$ 0.68 | 96.62 $\pm$ 4.69 |

**Note.** Random-Neg denotes the original corruption-based random negative sampling protocol. Hard-Neg@Top10 denotes the structure-based hard negative protocol, where the corrupted drug is preferentially replaced by one of its top-10 most similar drugs according to Morgan fingerprint Tanimoto similarity, while avoiding known positive triplets. The model was trained under the original Train-only protocol and evaluated with different negative sampling protocols. Since Random-Neg and Hard-Neg@Top10 use the same trained Train-only checkpoints and the same positive S1/S2 triplets, Recall remains unchanged across the two protocols; the degradation under Hard-Neg@Top10 mainly comes from increased false positives on structurally similar hard negatives. The Random-Neg results in this table were obtained by independently re-evaluating the same checkpoints with dynamically re-sampled random negatives, and therefore may slightly differ from the main Table 3 while preserving the same near-saturated trend.

TABLE S36: Negative-sampling similarity profile under the standard Random-Neg and Hard-Neg@Top10 protocols on DB. Results are averaged over three folds.

| Setting | Negative protocol | Mean Tanimoto | Median Tanimoto | Hard-source ratio (%) | Fallback ratio (%) | Avg. pair-graph edges | Shuffle AUC |
|---|---|---|---|---|---|---|---|
| S1 | Random-Neg | 0.0953 $\pm$ 0.0003 | 0.0927 $\pm$ 0.0005 | 0.00 $\pm$ 0.00 | 0.00 $\pm$ 0.00 | 221.80 $\pm$ 6.93 | 49.53 $\pm$ 0.53 |
| S1 | Hard-Neg@Top10 | 0.3201 $\pm$ 0.0057 | 0.2799 $\pm$ 0.0076 | 97.11 $\pm$ 0.85 | 2.89 $\pm$ 0.85 | 220.81 $\pm$ 8.68 | 49.64 $\pm$ 0.89 |
| S2 | Random-Neg | 0.0945 $\pm$ 0.0001 | 0.0921 $\pm$ 0.0000 | 0.00 $\pm$ 0.00 | 0.00 $\pm$ 0.00 | 218.80 $\pm$ 1.87 | 49.99 $\pm$ 0.09 |
| S2 | Hard-Neg@Top10 | 0.3149 $\pm$ 0.0020 | 0.2771 $\pm$ 0.0027 | 97.02 $\pm$ 0.44 | 2.98 $\pm$ 0.44 | 218.75 $\pm$ 3.91 | 49.94 $\pm$ 0.27 |

**Note.** Mean and median Tanimoto similarities are computed between the original drug selected for corruption and its replacement drug using Morgan fingerprints. 'Hard-source ratio' denotes the proportion of negative samples generated from the top-10 structurally similar candidate list. 'Fallback ratio' denotes the proportion of cases where no valid hard negative was available after filtering known positive triplets and a random negative was used instead. Shuffle AUC is obtained by randomly shuffling labels and is expected to be close to 50%.

TABLE S37: Comparison of motif construction variants used for CMMIG granularity analysis.

| **Variant** | **BRICS decomposition** | **Additional splitting rules** | **Intra-drug clique** merging | **Purpose** |
|---|---|---|---|---|
| BRICS-only | Remove RDKit BRICS bonds and use the connected components as motifs. | Not used. | Not used. | Coarser motif granularity; evaluates the effect of using only standard BRICS fragments. |
| BRICS+splitting | BRICS bond breaking is applied. | Ring/non-ring boundary splitting and high-degree non-ring atom separation are applied. | Not used; exact duplicate cliques are removed, but overlapping non-identical cliques are retained. | Finer motif granularity; evaluates the effect of auxiliary splitting without merging and exposes possible over-fragmentation. |
| BRICS+splitting+merging | BRICS bond breaking is applied. | Same as BRICS+splitting. | Overlapping cliques within the same molecule are iteratively merged until no overlap remains. | Default D3I-COMCF setting; aims to reduce over-fragmentation and recover chemically coherent motif units. |

**Note.** Ring/non-ring boundary splitting refers to splitting a two-atom clique that crosses a ring/non-ring boundary. High-degree non-ring atom separation refers to separating a non-ring atom with more than two neighbors and its adjacent atoms into singleton cliques before optional merging.

TABLE S38: Cold-start performance comparison of different motif construction variants under the Train-only protocol on DB (%). Results are reported as mean $\pm$ standard deviation over the first three folds.

| **Setting** | **Motif construction** | **ACC** | **AUC** | **F1-score** | **AP** | **Precision** | **Recall** |
|---|---|---|---|---|---|---|---|
| S1 | BRICS-only | 72.04 $\pm$ 21.90 | 99.82 $\pm$ 0.26 | 53.50 $\pm$ 37.90 | 99.84 $\pm$ 0.23 | 100.00 $\pm$ 0.00 | 44.08 $\pm$ 43.82 |
| S1 | BRICS+splitting | 71.70 $\pm$ 25.46 | 94.84 $\pm$ 7.48 | 49.13 $\pm$ 48.88 | 95.83 $\pm$ 6.11 | 99.99 $\pm$ 0.01 | 43.41 $\pm$ 50.93 |
| S1 | BRICS+splitting+merging | 97.17 $\pm$ 3.31 | 100.00 $\pm$ 0.00 | 97.01 $\pm$ 3.57 | 100.00 $\pm$ 0.00 | 100.00 $\pm$ 0.00 | 94.35 $\pm$ 6.61 |
| S2 | BRICS-only | 89.95 $\pm$ 7.32 | 99.99 $\pm$ 0.02 | 88.32 $\pm$ 9.22 | 99.99 $\pm$ 0.02 | 100.00 $\pm$ 0.00 | 79.89 $\pm$ 14.65 |
| S2 | BRICS+splitting | 80.27 $\pm$ 18.16 | 98.83 $\pm$ 1.23 | 71.26 $\pm$ 27.86 | 99.08 $\pm$ 1.00 | 99.99 $\pm$ 0.02 | 60.55 $\pm$ 36.32 |
| S2 | BRICS+splitting+merging | 98.29 $\pm$ 2.38 | 100.00 $\pm$ 0.00 | 98.22 $\pm$ 2.49 | 100.00 $\pm$ 0.00 | 100.00 $\pm$ 0.00 | 96.59 $\pm$ 4.75 |

**Note.**BRICS+splitting+merging is the default motif construction strategy used in D3I-COMCF. The first three folds are used for a fair comparison because BRICS-only and BRICS+splitting were evaluated on the same three folds.

TABLE S39: CMMIG granularity statistics of BRICS-only and BRICS+splitting under the Train-only protocol on DB (%). Results are reported as mean $\pm$ standard deviation over the first three folds.

| **Motif construction** | **Mol. nodes** | **Motif nodes** | **Mol.-motif edges** | **Avg. motifs/drug** | **Avg. atoms/motif** | **Median atoms/motif** | **Small motif ratio ($\leq$2)** | **Feature failures** |
|---|---|---|---|---|---|---|---|---|
| BRICS-only | 1346.33 $\pm$ 8.02 | 1267.33 $\pm$ 11.15 | 4103.00 $\pm$ 12.49 | 3.05 $\pm$ 0.03 | 15.38 $\pm$ 0.10 | 14.00 $\pm$ 0.00 | 0.03 $\pm$ 0.00 | 0.00 $\pm$ 0.00 |
| BRICS+splitting | 1346.33 $\pm$ 8.02 | 64.67 $\pm$ 2.52 | 8534.67 $\pm$ 18.18 | 6.34 $\pm$ 0.03 | 1.79 $\pm$ 0.01 | 2.00 $\pm$ 0.00 | 0.93 $\pm$ 0.01 | 37.00 $\pm$ 2.65 |
| BRICS+splitting+merging | 1346.33 $\pm$ 8.02 | 477.33 $\pm$ 9.50 | 6716.67 $\pm$ 11.06 | 4.99 $\pm$ 0.03 | 12.74 $\pm$ 0.14 | 12.00 $\pm$ 0.00 | 0.074 $\pm$ 0.003 | 0.00 $\pm$ 0.00 |

**Note.** 'Small motif ratio ($\leq$2)' denotes the proportion of motif types containing no more than two atoms. 'Feature failures' denotes the number of motifs for which RDKit failed to generate a valid submolecule, sanitize the fragment, or compute the six motif-level physicochemical descriptors. In such cases, a zero vector was used as the motif feature. BRICS+splitting produces more motif instances per drug but most motif types become tiny fragments, indicating severe over-fragmentation.

TABLE S40: Ablation study on retaining only occurrence-frequency molecule-motif weights in CMMIG on DeepDDI (%).

| **Model** | **ACC** | **AUC** | **F1-score** | **AP** | **Precision** | **Recall** |
|---|---|---|---|---|---|---|
| D3I-COMCF | 95.63 $\pm$ 0.16 | 99.10 $\pm$ 0.07 | 97.06 $\pm$ 0.11 | 99.67 $\pm$ 0.02 | 98.68 $\pm$ 0.08 | 95.49 $\pm$ 0.20 |
| Occurrence-only | 94.97 $\pm$ 0.61 | 98.97 $\pm$ 0.29 | 96.60 $\pm$ 0.42 | 99.62 $\pm$ 0.12 | 98.64 $\pm$ 0.19 | 94.65 $\pm$ 0.65 |

**Note.** In Occurrence-only, the merged CMMIG structure, motif definition, motif features, and RBF edge encoding are kept unchanged. Only the TF-IDF-style rare-motif-aware reweighting factor is removed, so the molecule-motif edge weight retains the normalized motif occurrence frequency within each drug.

## B. Supplementary figure

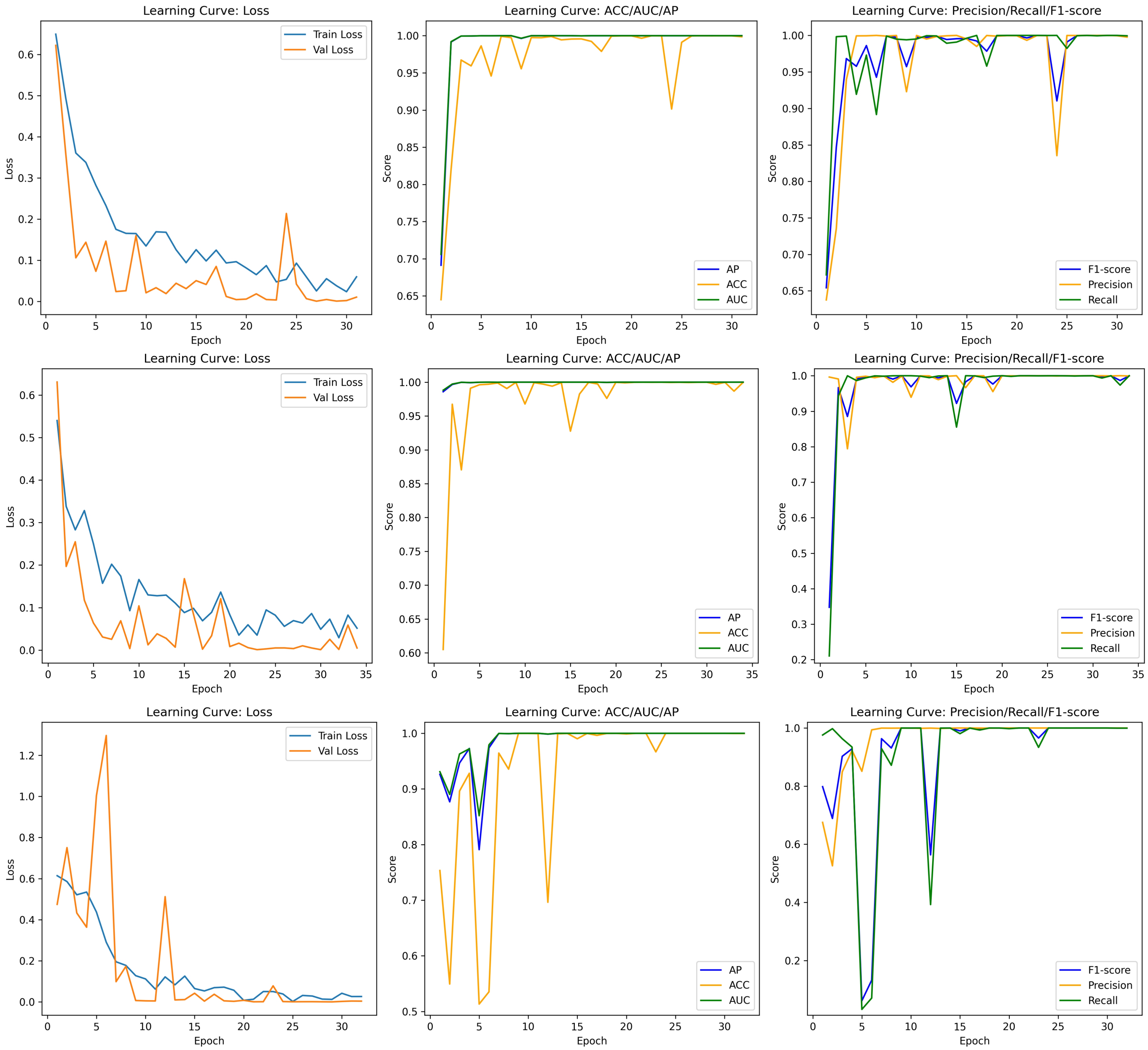



Fig. S1: Training/validation learning curves (loss, AP, ACC, AUC, F1-score, Precision, and Recall) on the DB dataset for three folds.

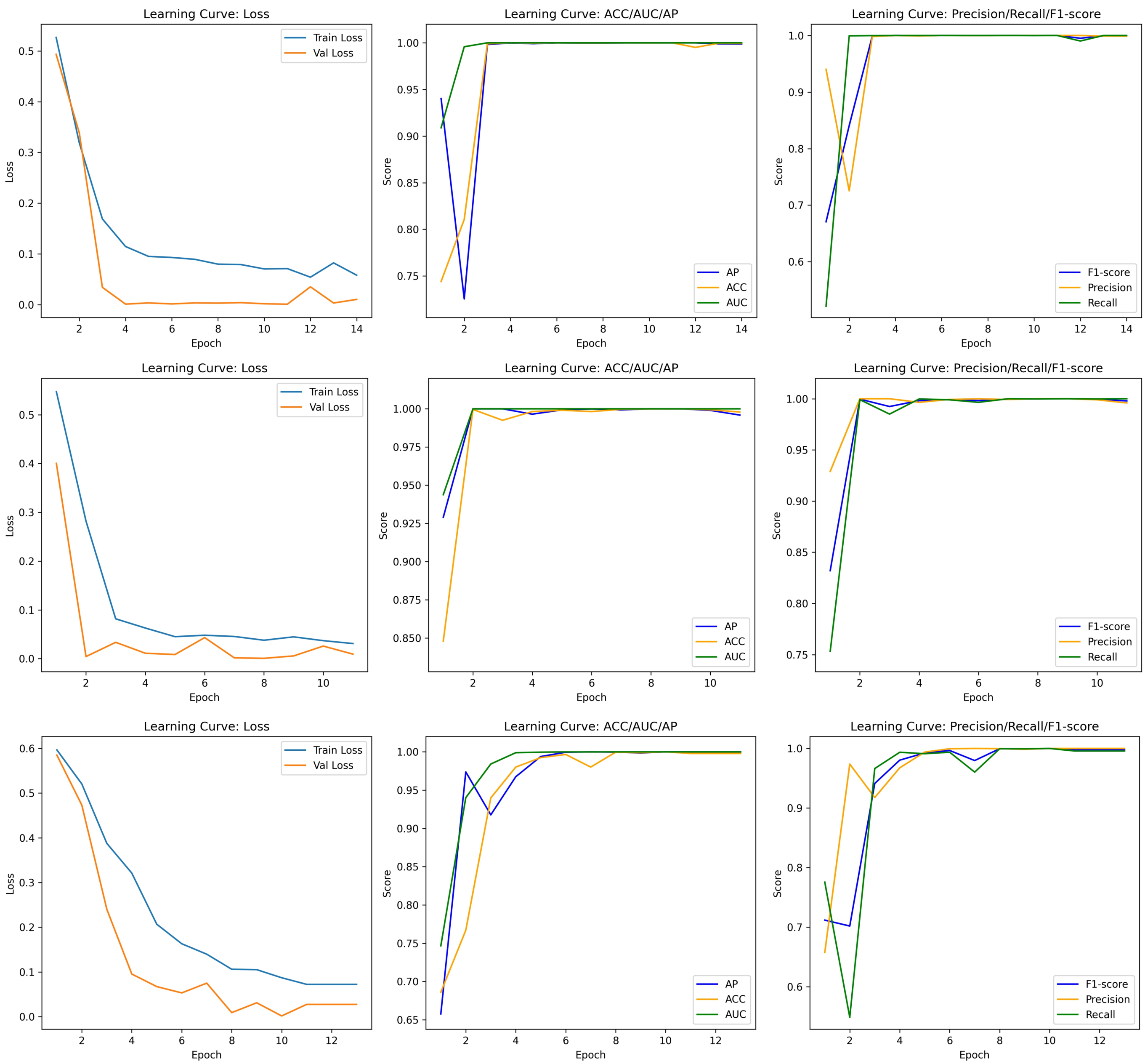



Fig. S2: Training/validation learning curves (loss, AP, ACC, AUC, F1-score, Precision, and Recall) on the Twosides dataset for three folds.

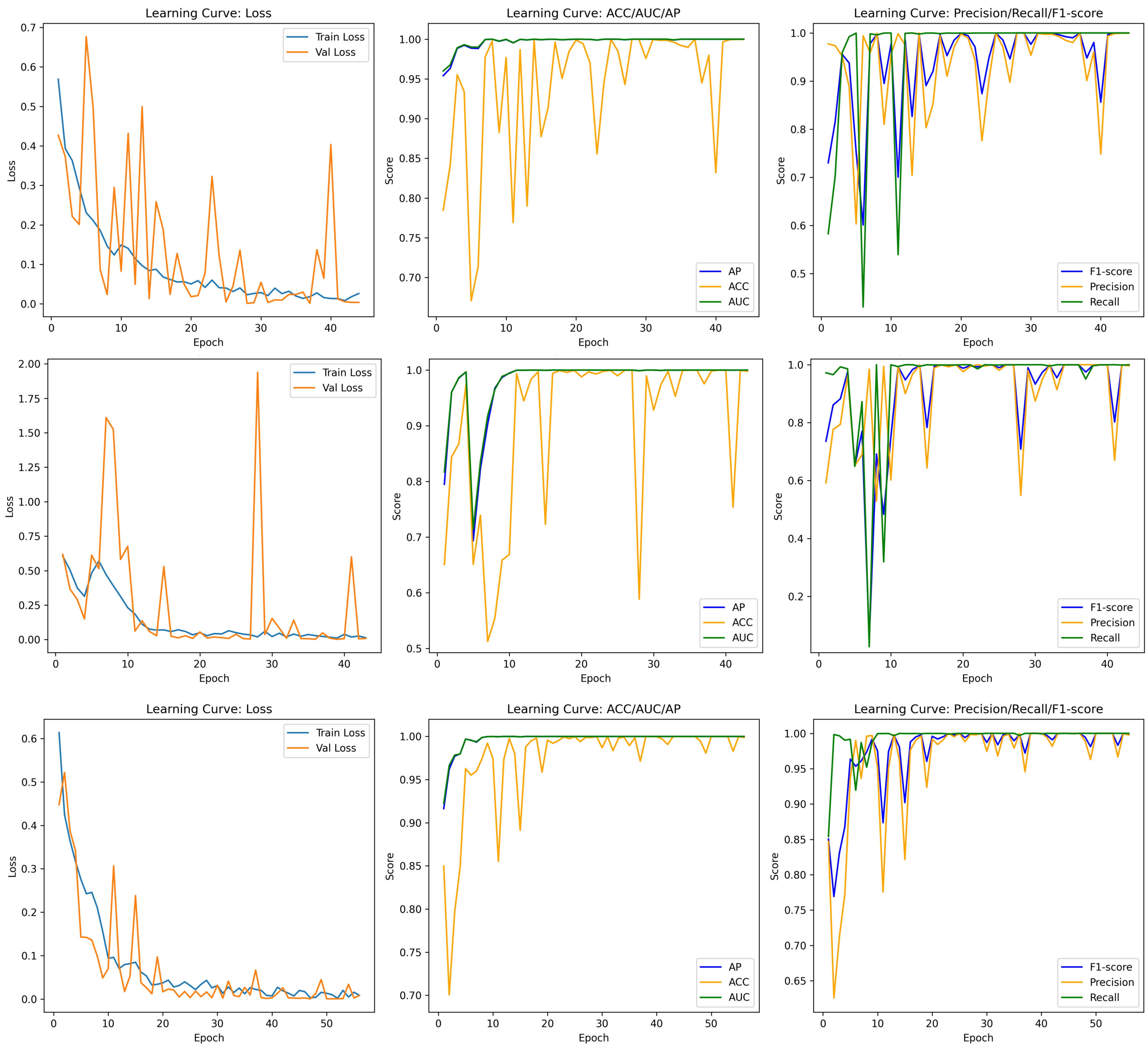



Fig. S3: Training/validation learning curves (loss, AP, ACC, AUC, F1-score, Precision, and Recall) under the Train-only protocol (cold-start scenario) for three folds.

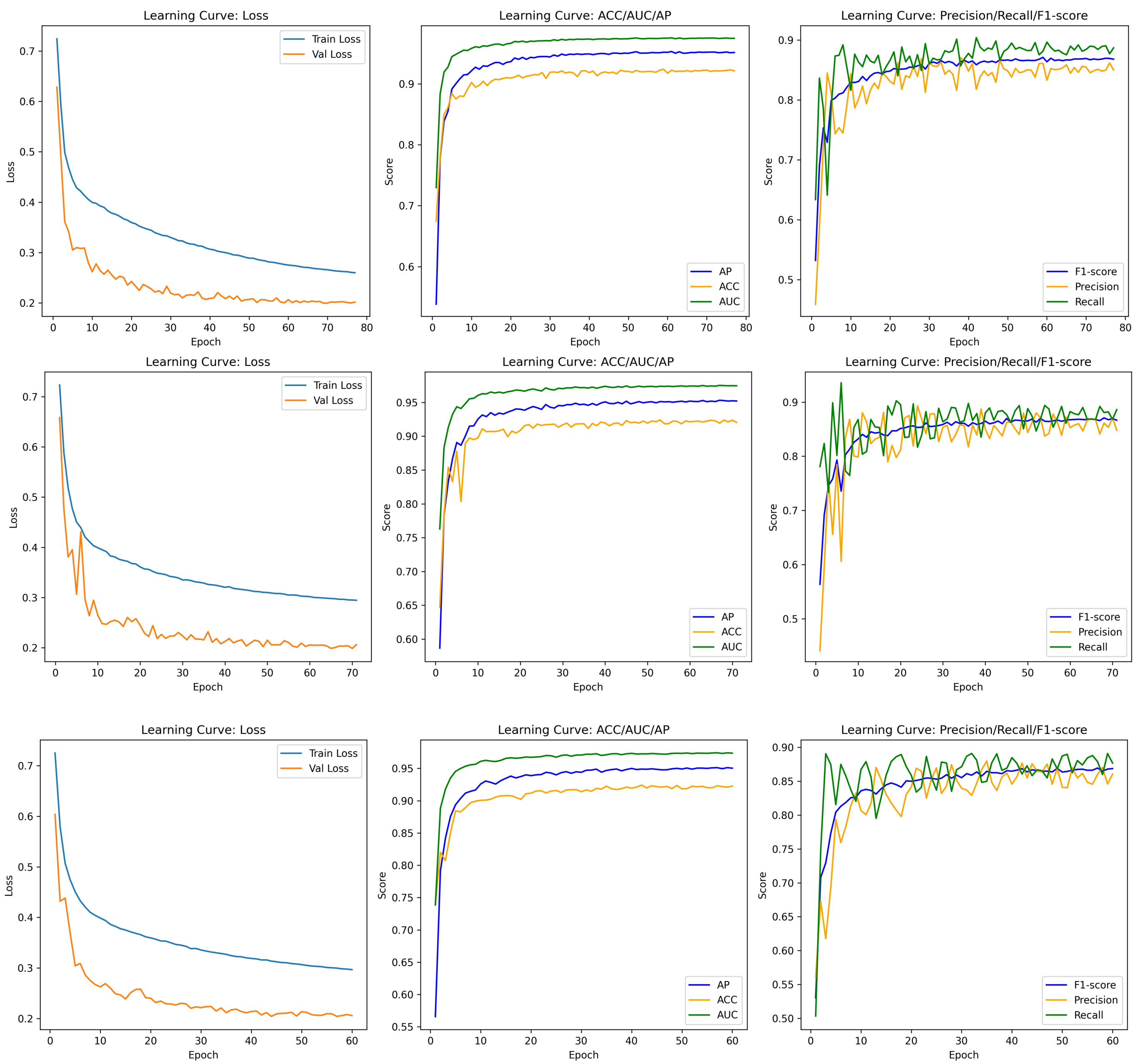



Fig. S4: Training/validation learning curves (loss, AP, ACC, AUC, F1-score, Precision, and Recall) on the ZhangDDI dataset for three folds.

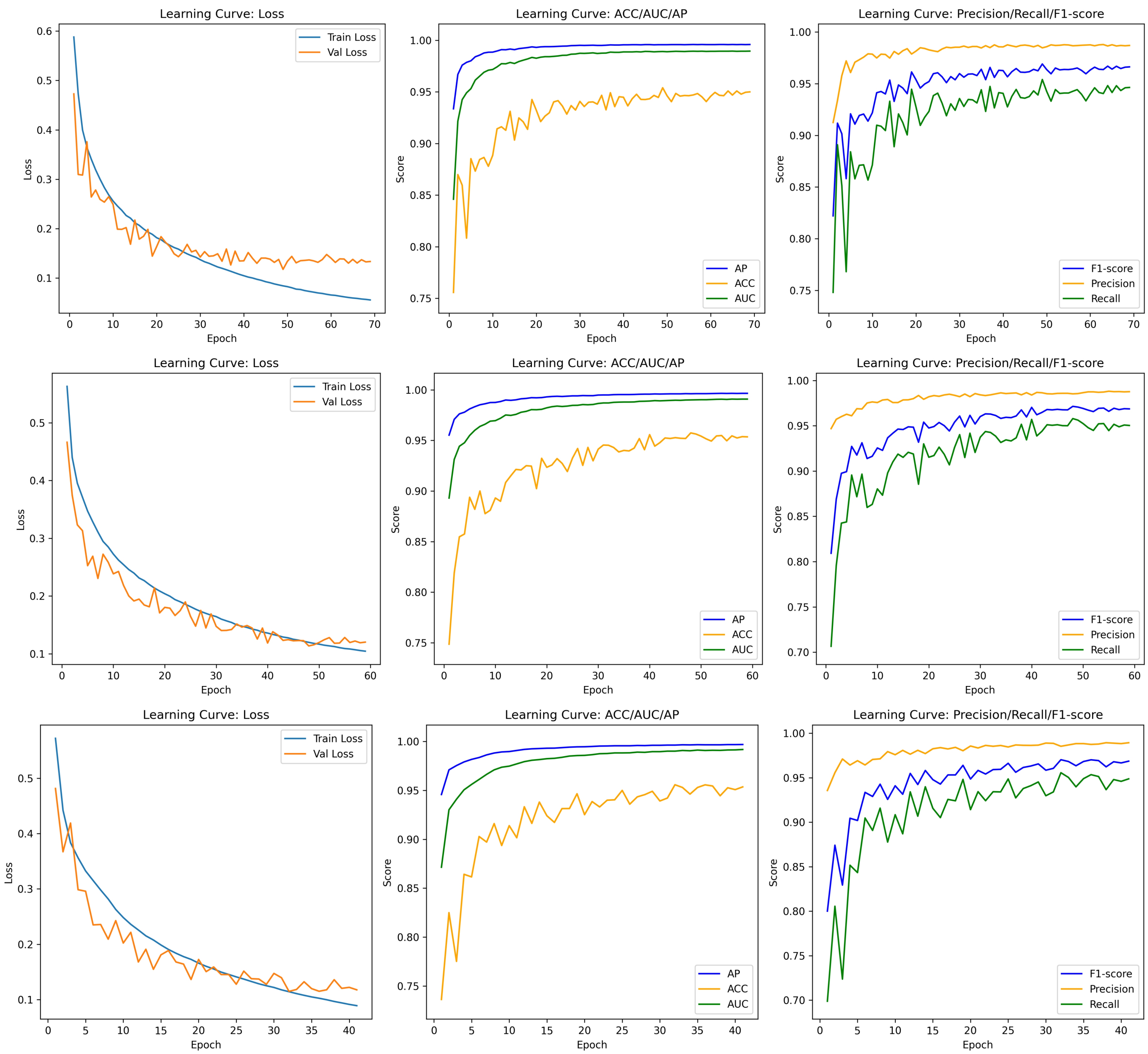
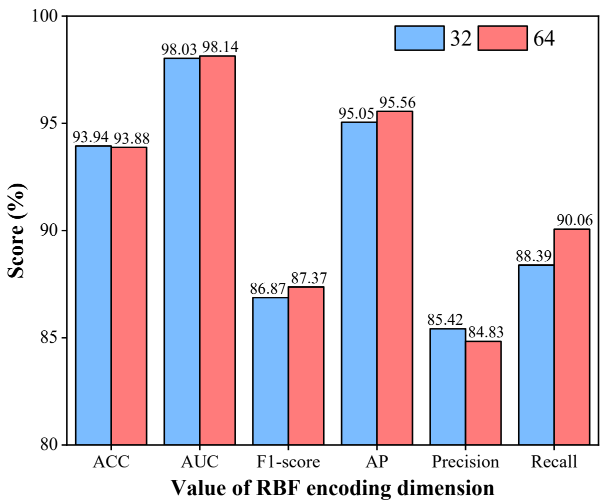
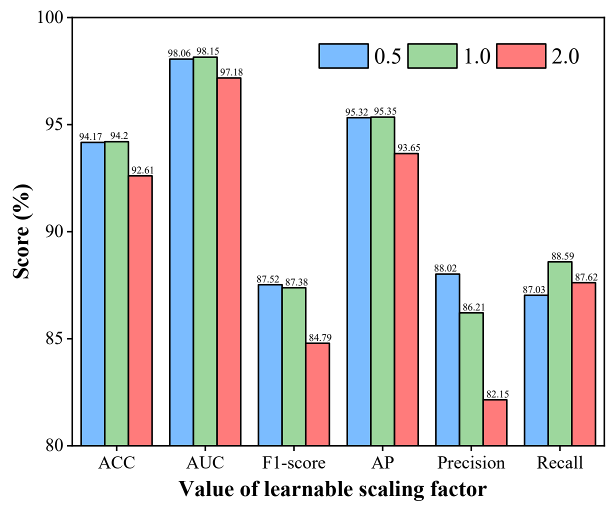



Fig. S5: Training/validation learning curves (loss, AP, ACC, AUC, F1-score, Precision, and Recall) on the DeepDDI dataset for three folds.

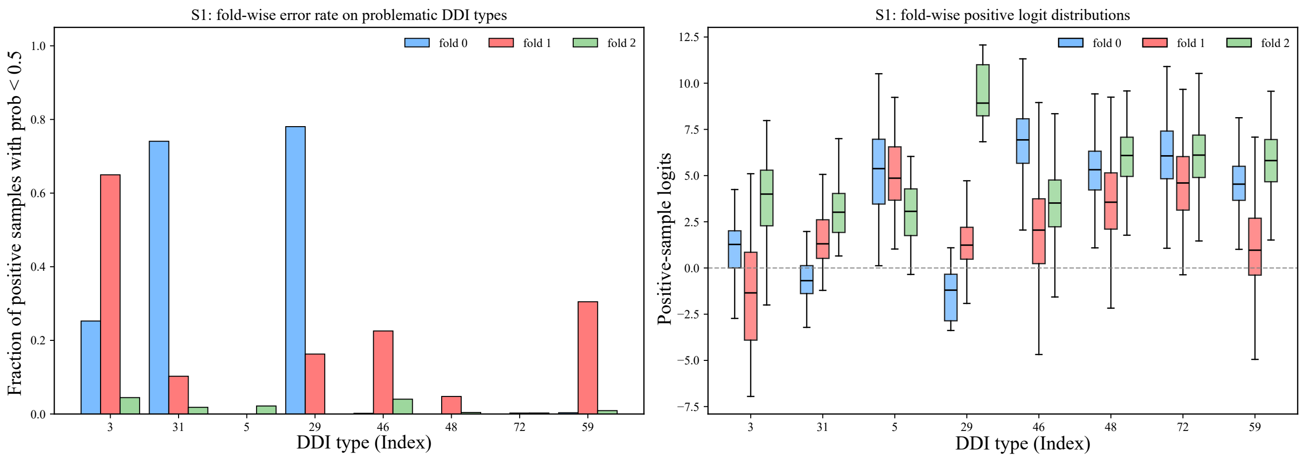



(a): Sensitivity to the RBF encoding dimension $K$ on a representative ZhangDDI fold.

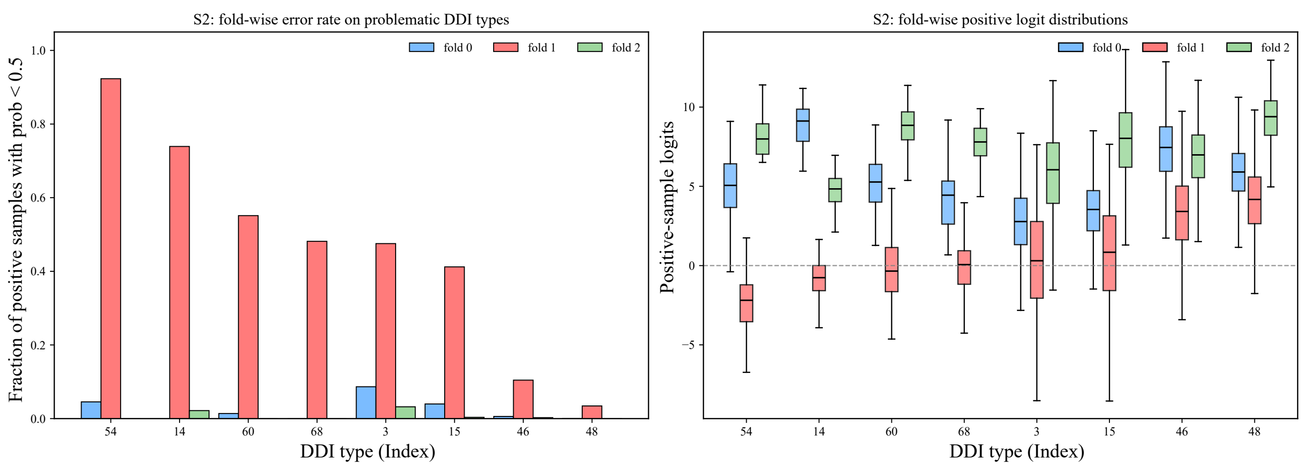



(b): Sensitivity to the learnable scaling factor $\gamma$ on a representative ZhangDDI fold.

Fig. S6: Parameter sensitivity analysis on ZhangDDI. (a) Effect of the RBF encoding dimension $K$. (b) Effect of the learnable scaling factor $\gamma$. The results support the use of $\gamma = 1.0$ and $K = 32$ in the main experiments, as these settings provide the best overall balance across the evaluation metrics.

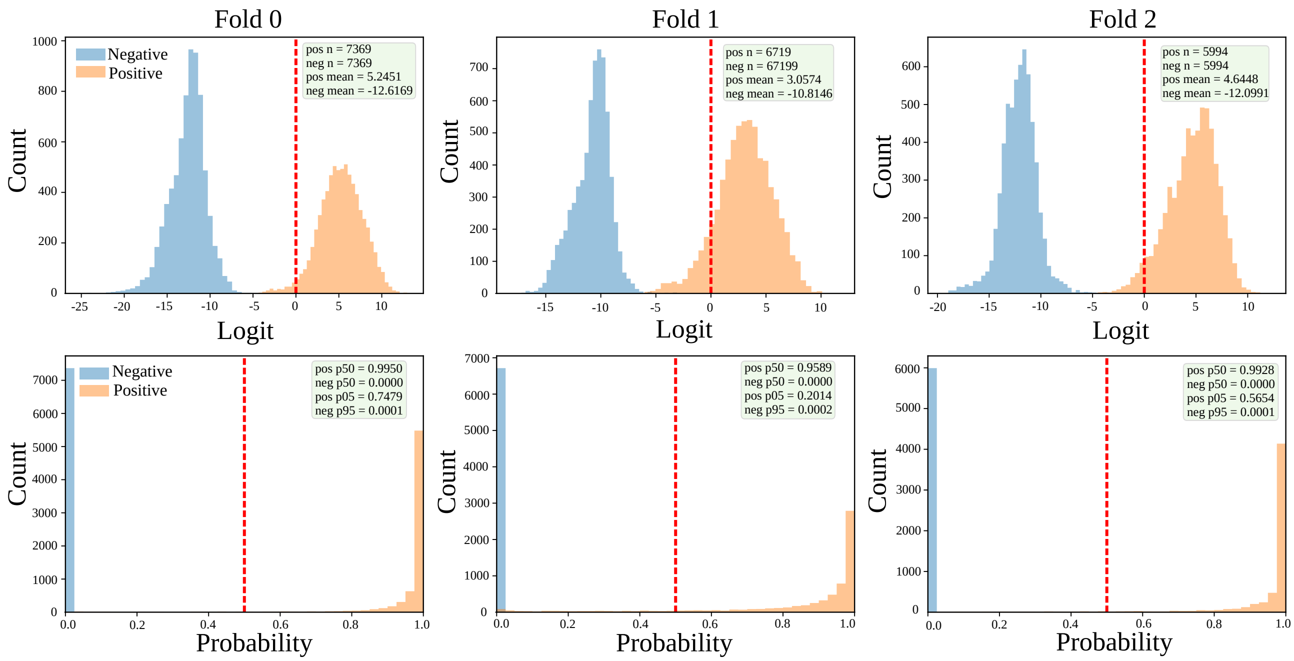



(a) S1: fold-wise comparison for representative DDI types showing fold1-specific degradation.

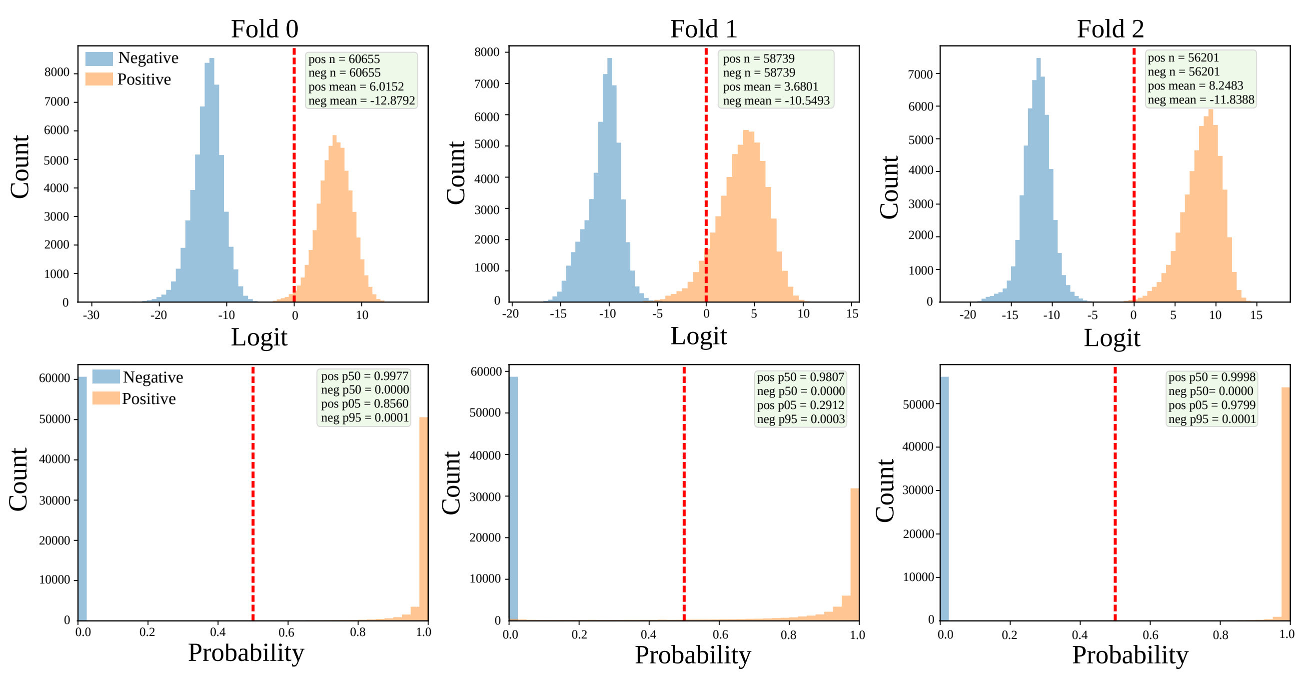



(b) S2: fold-wise comparison for representative DDI types showing fold1-specific degradation.

Fig. S7: Fold-wise relation-specific score distributions under the Train-only protocol.




(a) S1 (new drug, new drug): fold-wise overlap distributions of positive and negative logits and probabilities.




(b) S2 (new drug, existing drug): fold-wise overlap distributions of positive and negative logits and probabilities.

Fig. S8: Fold-wise overlap distributions of positive and negative prediction scores under the Train-only protocol. In each subfigure, the upper row shows the logit distributions and the lower row shows the probability distributions across folds.

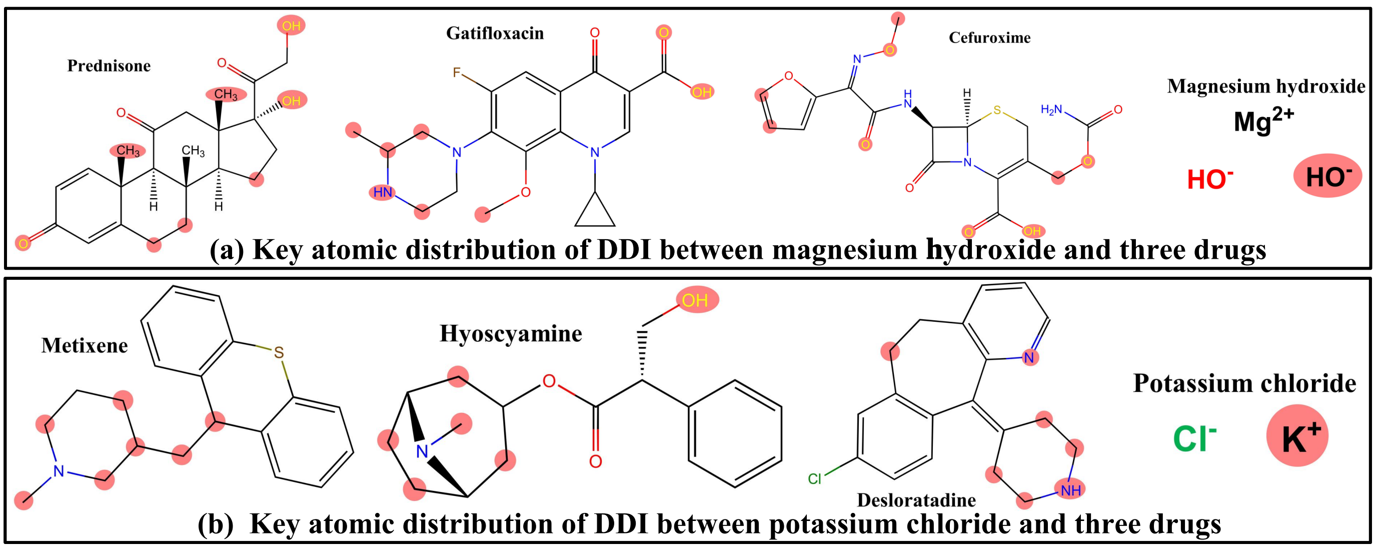



Fig. S9: Atom-level DDI interpretation for representative drugs that cannot be effectively decomposed into BRICS motifs. (a) Key atoms for magnesium hydroxide paired with prednisone, gatifloxacin, and cefuroxime (DDI types: bioavailability decrease, absorption decrease, and serum concentration decrease, respectively). (b) Key atoms for potassium chloride paired with metixene, hyoscyamine, and desloratadine (all labeled as ulcerogenic activities Chloride increase). Key atoms are obtained by aggregating SAGPooling and readout attention across AF-LDR blocks, highlighting the top 30% atoms.

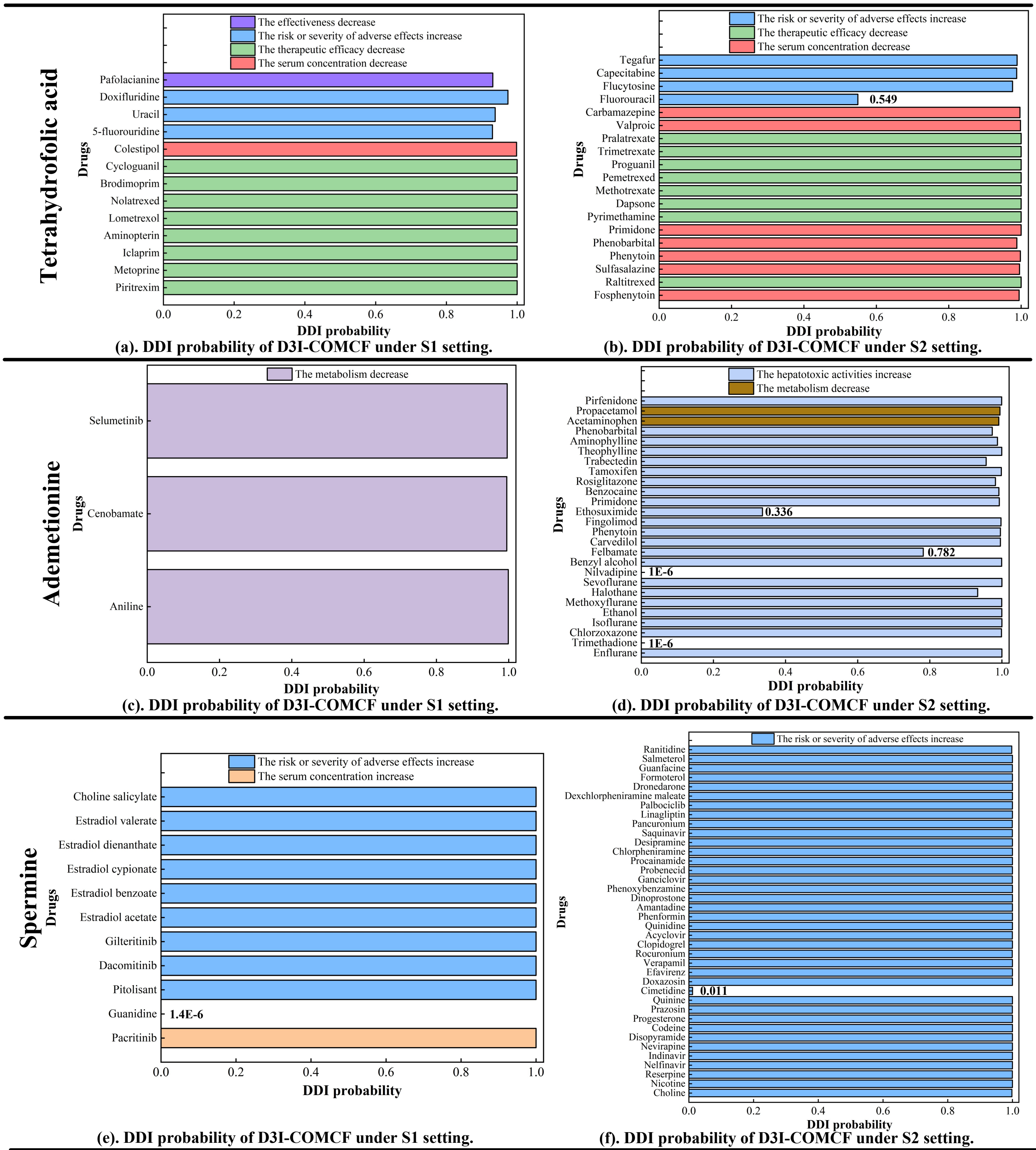
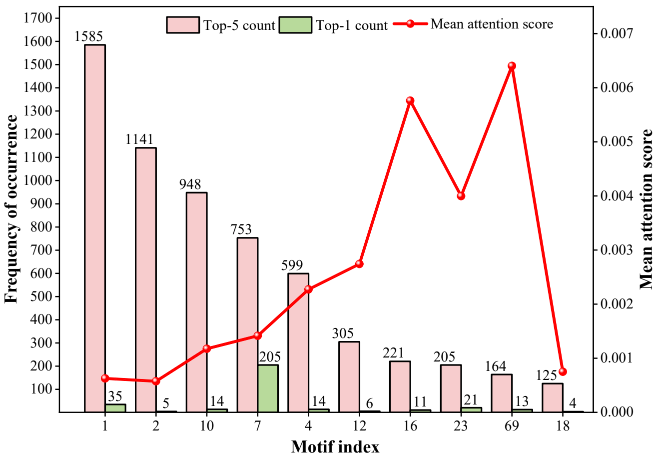
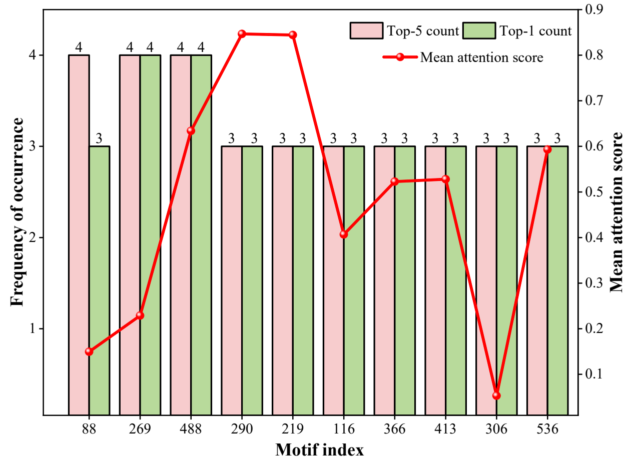


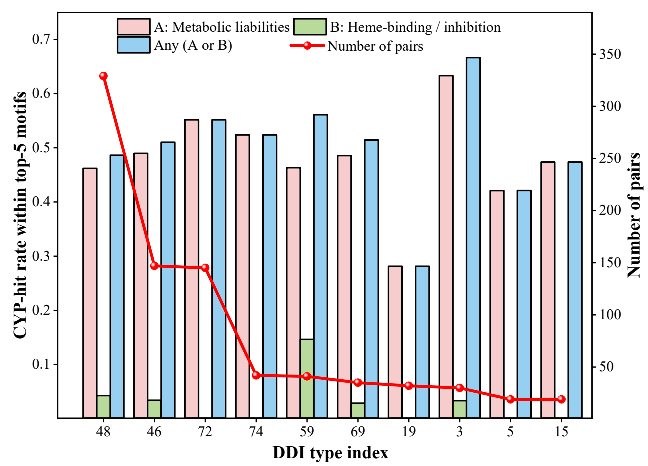

Fig. S10: Case-oriented visualization of the cold-start generalization behavior of D3I-COMCF for three newly introduced drugs (Tetrahydrofolic acid, Ademetionine, and Spermine) under S1 (new-new) and S2 (new-existing) settings. The six panels are organized by drug and setting to preserve the correspondence between each newly introduced drug, its candidate interaction partners, and the associated DDI types. In total, 110 candidate pairs were constructed (32, 29, and 49), covering eight DDI types. Predictions are nearly saturated (≈1.0) for almost all pairs. Only four near-zero outliers are observed: Ademetionine-Trimethadione (S2, 1E-6), Ademetionine-Nilvadipine (S2, 1E-6), Spermine-Guanidine (S1, 1.4E-6), and Spermine-Cimetidine (S2, 0.0113).



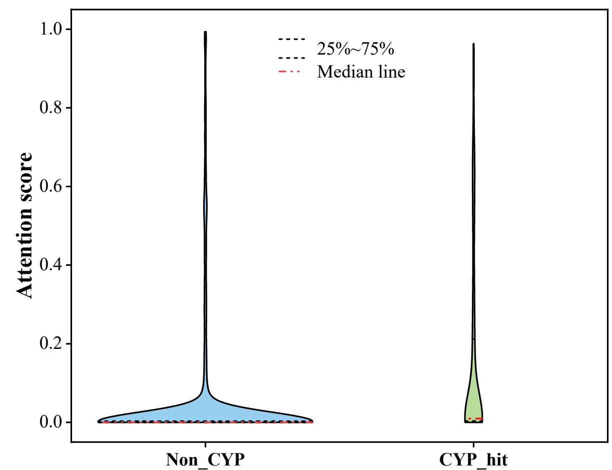

(a): The top 10 most frequently highlighted motifs.



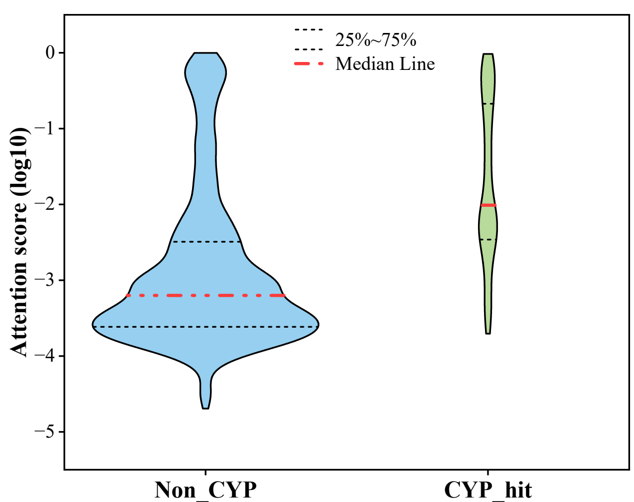

(b): The bottom 10 least frequently highlighted motifs.




(c): CYP-hit rate within top-5 motifs across major DDI types.




(d): Violin plot of raw attention scores.




(e): Violin plot of log10-transformed attention scores.

Fig. S11: Systematic interpretability statistics on the top 1,000 drug pairs sampled from the DrugBank fold1 test set. (a) Frequency distribution of the most frequently highlighted motifs, showing that a small set of motifs accounts for a large proportion of highlighted occurrences. (b) Representative low-frequency motifs and their mean attention scores, indicating that some rare motifs can still receive high importance. (c) CYP-hit rates across DDI types based on the manually curated SMARTS alert library, including category A (metabolic liabilities), category B (heme-binding/inhibition-related motifs), and their union. (d) Violin plot of raw attention scores for CYP-hit and non-CYP motif occurrences. (e) Violin plot of log10-transformed attention scores, which improves the visualization of the non-CYP distribution concentrated near zero.